

Project Report

Wazuh Local Lab Monitoring & Alerting

Submitted by: Mohammed Ifaz M

University / Institution: Blearn Academy

Date: 16 July 2025

Project Guide / Supervisor: Gladwin M.S

Index

S. No	Topic	Page
1	Introduction	3
2	Lab Architecture & Setup	3
3	Installing & Configuring Wazuh	4
4	Setting up the Windows Agent	5
5	Enabling SSH & Opening Port 22	6
6	Attack Simulation & Alerts	7
7	Results & Observations	8
8	References & Tools	9
9	Conclusion	10
10	Reflections & Learnings	10

1. Introduction

Project goal

The aim of this project was to build a local SOC (Security Operations Center) lab to understand, hands-on, how real-time security monitoring and alerting work.

Instead of just reading about SIEM tools, I wanted to test them in practice: collecting logs, simulating attacks, and observing how alerts appear in real-time — just like in professional cybersecurity operations.

By doing this, I learned how Wazuh, an open-source SIEM, helps detect suspicious activity, generate alerts, and visualize threats so security teams can respond faster.

2. Lab Architecture & Setup

To build this project, I created a small local network on my laptop using VirtualBox. I combined virtual machines with my actual Windows computer to make the lab environment feel realistic, just like a real office network.

Here's what each part of the lab did:

- **Wazuh Manager (Ubuntu VM)**
 - A virtual machine running Ubuntu inside VirtualBox
 - Installed Wazuh manager, Filebeat, and the Wazuh dashboard
 - Worked as the SIEM core: it collected logs from agents, analyzed them, and generated alerts
- **Windows 11 machine (my real PC)**
 - This was my actual physical Windows computer, not a VM
 - Installed the Wazuh agent to monitor logs, events, and security data
 - Sent this data to the Wazuh manager for analysis
- **Kali Linux VM**
 - Another VM running in VirtualBox
 - Used as an attacker machine to run tools like Nmap (for scanning open ports) and Hydra (for brute-force login attempts) targeting the Windows machine

All three systems were set to Bridged Adapter mode in VirtualBox. This put them on the same LAN (local area network), so:

- The Kali VM could see and scan the Windows machine's open ports

- The Windows agent could talk directly to the Wazuh manager

3. Installing & Configuring Wazuh

On the Ubuntu VM (Wazuh Manager), I:

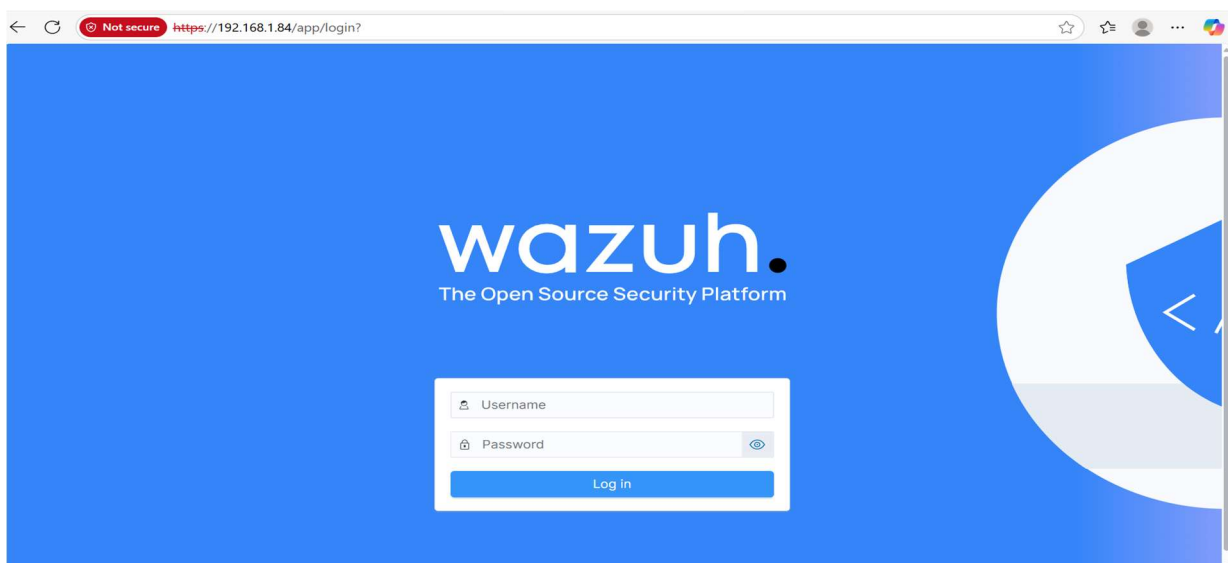
- Followed official Wazuh documentation
- Installed:
 - Wazuh manager (handles analysis & rules)
 - Filebeat (ships data to the dashboard)
 - Wazuh dashboard (web UI to view alerts)
- Configured services to start on boot
- Opened the dashboard in browser:

`https://<manager_ip>:5601`

- Made sure the dashboard correctly displayed connected agents and live alerts

✂ What I did:

- ☒ Installed Wazuh manager
- ☒ Configured Filebeat
- ☒ Tested the dashboard
- ☒ Verified everything worked end-to-end



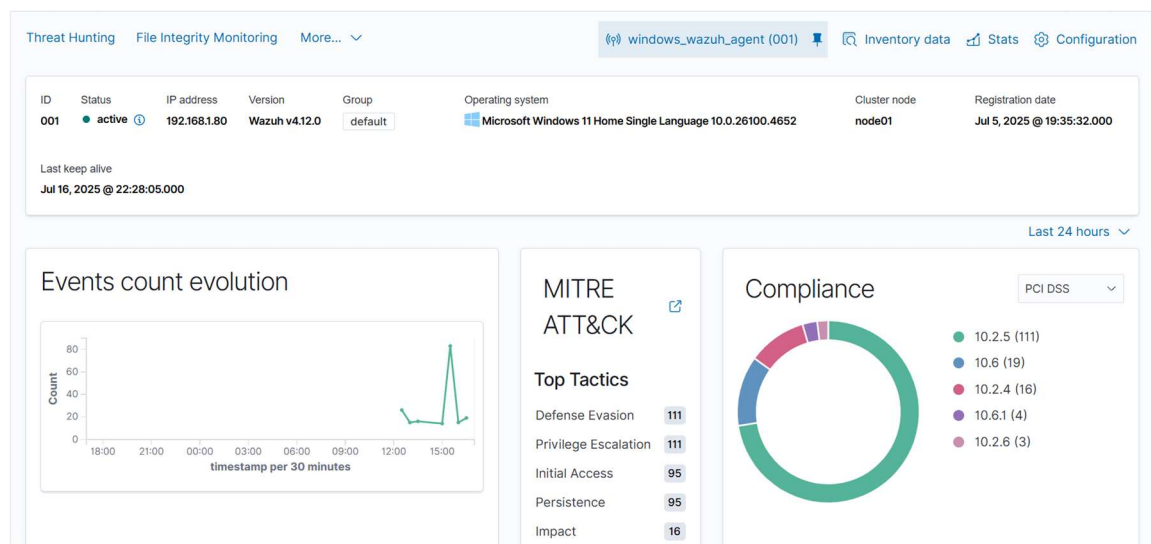
4. Setting up the Windows Agent

On my physical Windows 11 system (not a VM):

- Downloaded and installed the **Wazuh agent**
- Configured it with the Wazuh manager IP address
- Exchanged keys to authorize the agent
- Checked from the Wazuh dashboard that the agent showed as “active”

📌 What I did:

- ✓ Installed agent
- ✓ Connected to Wazuh manager
- ✓ Confirmed it's actively sending data



5. Enabling SSH & Opening Port 22

To simulate attacks, I needed SSH running on my Windows machine.

- Installed OpenSSH Server:
Add-Windows Capability -Online -Name OpenSSH.Server
- Started the SSH service:
Start-Service sshd
- Checked if port 22 was listening:
netstat -an | findstr :22
- Turned off Windows firewall to let Kali VM scan the port:
Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False

📌 What I did:

- ✓ Installed OpenSSH Server
- ✓ Confirmed SSH is running
- ✓ Turned off firewall
- ✓ Verified port 22

```
Administrator: Windows PowerShell

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

$ C:\WINDOWS\system32> Restart-Service sshd
$ C:\WINDOWS\system32> netstat -an | findstr :22
TCP    0.0.0.0:22          0.0.0.0:0           LISTENING
TCP    [::]:22            [::]:0              LISTENING
$ C:\WINDOWS\system32> Set-NetFirewallProfile -Profile Domain,Public,Private -Enabled False
```

6. Attack Simulation & Alerts

a) Port scan: From Kali VM, ran:

```
nmap -p 22 windows_ip
```

b) SSH brute-force: Used hydra:

```
hydra -l admin -P /usr/share/wordlists/rockyou.txt ssh://windows_ip
```

c) Login failures: Wazuh collected Windows login failure events and sent alerts.

🔴 What I did:

- ✅ Ran Nmap scan
- ✅ Tried brute-force with hydra
- ✅ Watched alerts appear in dashboard

```
(kali㉿kali)-[~]
$ nmap -sS 192.168.29.48 -p 22
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-16 00:43 EDT
Nmap scan report for 192.168.29.48
Host is up (0.00065s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
MAC Address: 5C:BA:EF:AB:39:25 (Chongqing Fugui Electronics)

Nmap done: 1 IP address (1 host up) scanned in 0.48 seconds

(kali㉿kali)-[~]
$ nmap -sS 192.168.29.48 -p 22
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-16 00:46 EDT
Nmap scan report for 192.168.29.48
Host is up (0.00030s latency).

PORT      STATE      SERVICE
22/tcp    open      ssh
MAC Address: 5C:BA:EF:AB:39:25 (Chongqing Fugui Electronics)

Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds

(kali㉿kali)-[~]
$ nmap -sS 192.168.29.48 -p 22
Starting Nmap 7.95 ( https://nmap.org ) at 2025-07-16 00:
Nmap scan report for 192.168.29.48
Host is up (0.00030s latency).

PORT      STATE      SERVICE
22/tcp    open      ssh
```

```

(kali@kali)-[~]
$ ssh Administrator@192.168.29.48
The authenticity of host '192.168.29.48 (192.168.29.48)' can't be established.
ED25519 key fingerprint is SHA256:InlmKY8QIfQ3qfRYtboJuk0ipP6vcMAa93G/sFgUU0.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.29.48' (ED25519) to the list of known hosts.
Connection reset by 192.168.29.48 port 22

(kali@kali)-[~]
$ hydra -l molly -P pass.txt ssh://192.168.29.48
hydra v9.5 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal
hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-07-16 00:46:23
WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
DATA] max 14 tasks per 1 server, overall 14 tasks, 14 login tries (l:1/p:14), ~1 try per task
DATA] attacking ssh://192.168.29.48:22/
of 1 target completed, 0 valid password found
hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-07-16 00:46:24

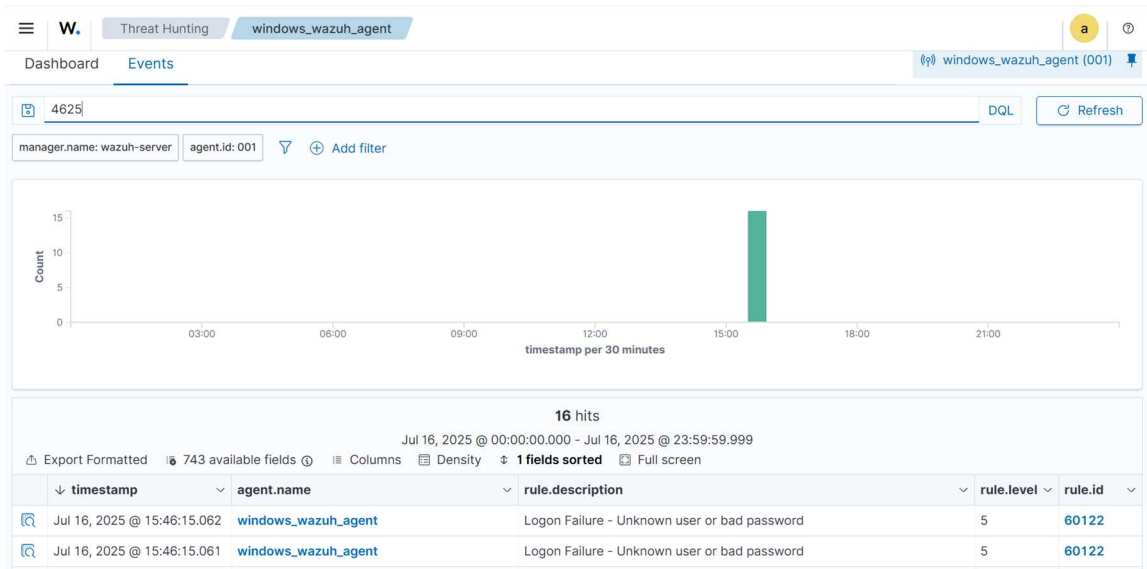
```

7. Results & Observations

- Wazuh agent captured port scans, brute-force attempts, and login failures
- Alerts appeared in real-time on the dashboard
- Disabling firewall made attacks easier to test, but it's risky in production

📌 What I noticed:

- ✅ Wazuh detects brute-force and scans
- ✅ Dashboard updates quickly
- ✅ Firewall off helped for lab testing



16 hits

Jul 16, 2025 @ 00:00:00.000 - Jul 16, 2025 @ 23:59:59.999

Export Formatted 743 available fields Columns Density 1 fields sorted Full screen

timestamp	agent.name	rule.description	rule.level	rule.id
 Jul 16, 2025 @ 15:46:15.062	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.061	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.055	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.054	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.034	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.020	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:15.002	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.991	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.962	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.957	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.942	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.919	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:14.919	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122
 Jul 16, 2025 @ 15:46:06.851	windows_wazuh_agent	Logon Failure - Unknown user or bad password	5	60122

8. References & Tools

- Wazuh docs
- VirtualBox
- Windows 11
- Kali Linux
- Hydra
- Nmap

9. Conclusion

By setting up virtual machines alongside my real Windows computer, I created a small but realistic SOC lab on my laptop. In this lab, my Windows machine acted as the monitored system, the Ubuntu VM ran Wazuh to collect and analyze logs, and the Kali Linux VM simulated attacks like port scans and brute-force attempts. Because everything was on the same network, Wazuh could see real events as they happened. When I ran attacks from Kali, the Wazuh agent on Windows sent logs to the manager, which raised alerts visible on the dashboard. This hands-on setup helped me clearly see how SOC teams detect and respond to threats in real time.

10. Reflections & Learnings

Through this project, I didn't just learn about tools — I learned how security teams actually work in practice. By setting up my own SOC lab, I could see how logs, events, and alerts come together to tell the story of what's really happening on a network.

I also understood the importance of:

- Properly configuring agents and manager systems
- Using open-source tools like Wazuh for cost-effective monitoring
- Testing with real attack scenarios to verify detection works
- Keeping security layers (like firewalls) and why turning them off should only be done carefully in a lab

Overall, this project showed me that monitoring and alerting aren't just about collecting data — they're about making sense of data, spotting patterns, and reacting quickly to protect systems.